

BUSINESS VALUE REVIEW

Harness IaCM — Business Value Review

Infrastructure as Code Management · Adoption & Maturity Assessment

Account: SxuV0ChbRqWGSYCIFIMQMQ

May 8, 2026

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Generated: May 8, 2026 | Account ID: SxuV0ChbRqWGSYCIFIMQMQ

Executive Summary

INFO

This Business Value Review assesses the current adoption, governance posture, and feature utilisation of Harness Infrastructure as Code Management (IaCM) across the entire account. Data was collected live from the Harness platform via the IaCM MCP Server.

Key Findings:

- The account has 22 workspaces and 15 IaCM pipelines spread across 2 orgs and 9 projects — a solid foundation indicating active IaCM usage.
- OPA governance is fully deployed — all 15 pipelines (100%) are covered by account-level policy sets, showing a mature policy framework.
- Significant feature gaps exist — Checkov security scanning (13%), cost estimation (0%), and IaCM templates (0%) represent the largest expansion opportunities.
- 46 OPA policies are authored but only 2 of 6 policy sets are enabled — 4 policy sets containing S3 compliance rules are written but inactive.

Section I — IaCM Footprint

1.1 Account Inventory

Org	Project	Workspaces	Pipelines	Status
`default`	gketest	19	10	Active
`default`	chaos_project	3	4	Active
`default`	trainingtest	0	0	Configured, unused
`default`	chaos_test_project1	0	0	Configured, unused
`default`	LinuxDemo	0	0	Configured, unused
`default`	projectx	0	0	Configured, unused
`sam`	CCM-Demo	0	1	Pipeline only
`sam`	fftest1	0	0	Configured, unused
`sam`	fftestqa	0	0	Configured, unused

WARNING

7 of 9 IaCM-enabled projects have zero workspaces. The IaCM module has been enabled but not utilised in these projects. This represents untapped expansion within the existing account footprint.

1.2 Workspace Breakdown — `gketest` (primary project)

Environment	Count	Provisioner
DEV	7	Terraform

QA	6	Terraform
PROD	3	Terraform
OpenTofu	3	OpenTofu
Total	**19**	—

INFO

The naming convention (DEV-*, QA-*, PROD-*) indicates a structured environment promotion pattern — a positive signal of IaCM maturity. Several workspaces have a status: failed indicating active operational issues to address.

1.3 Workspace Status Distribution

Status	Count	%
`active`	14	64%
`apply_needed`	4	18%
`failed`	4	18%

RISK

4 workspaces (18%) are in failed status and 4 are apply_needed. These may indicate drift, broken connectors, or stalled runs that need remediation.

Section II — Pipeline & Feature Adoption

2.1 IaCM Pipelines

Project	Pipeline	Purpose
gketest	Jira EC2 Provisioning	EC2 provisioning via Jira automation
gketest	Jira S3 Provisioning	S3 provisioning via Jira automation
gketest	IaCM Jira Automation	Base Jira IaCM integration
gketest	create-s3-site	S3 static site creation
gketest	s3-dev-apply	S3 dev environment provisioning
gketest	s3-dev-destroy	S3 dev environment teardown
gketest	s3-prod-apply	S3 production provisioning
gketest	s3-dev-drift	S3 drift detection
gketest	tofus3-deploy	OpenTofu S3 deployment
gketest	tofus3-destroy	OpenTofu S3 teardown
chaos_project	iacm-provision-pipeline	General IaCM provisioning
chaos_project	iamcm-destroy-pipeline	IaCM teardown
chaos_project	ccm_connector_creation	CCM connector automation
chaos_project	Demo_Pipeline	Demo pipeline
CCM-Demo	Selective Approval Notification	Approval workflow

INFO

The account has a mature Jira-to-IaCM automation pattern (3 pipelines), showing advanced integration capability. Drift detection (s3-dev-drift) is also present — an advanced IaCM practice.

2.2 Feature Adoption Scorecard

Feature	Scope	Adopted	Total	%	Priority
Checkov Security Scans	Pipelines	2	15	13%	High
Cost Estimation	Workspaces	0	22	0%	High
IaCM Templates	Pipelines	0	15	0%	Medium
Private Module Registry	Workspaces	0	22	0%	Low-Medium

2.3 Checkov Security Scans

CRITICAL

Only 2 of 15 pipelines (13%) run Checkov security scans.

Pipelines running Checkov: Jira EC2 Provisioning, s3-dev-apply

Pipelines WITHOUT Checkov (13):

Jira S3 Provisioning, IaCM Jira Automation, tofus3-deploy, tofus3-destroy, s3-dev-destroy, s3-prod-apply, s3-dev-drift, create-s3-site, iacm-provision-pipeline, iamcm-destroy-pipeline, Demo_Pipeline, ccm_connector_creation, Selective Approval Notification Pipeline

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What this means: 87% of provisioning pipelines have no policy-as-code security scan before or after infrastructure changes. Misconfigurations (public S3 buckets, open security groups, missing encryption) can be provisioned without any automated detection.

Recommendation: Enable IaCMCheckov step in all provisioning pipelines. Prioritise s3-prod-apply, tofus3-deploy, and s3-dev-destroy first.

2.4 Cost Estimation

CRITICAL

Cost estimation is disabled on all 22 workspaces (0% adoption).

Without cost estimation, teams have no visibility into the financial impact of a Terraform plan before it is applied. Infrastructure spend changes are discovered after the fact — not before.

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Recommendation: Enable `cost_estimation_enabled: true` on all production and QA workspaces. Connect with Harness CCM to surface pre-apply cost diffs in the pipeline UI and OPA cost guardrails (the Terraform Plan Cost - Total Cost Estimate policy already exists — it just needs to be activated).

2.5 IaCM Templates

WARNING

Zero pipelines use IaCM templates (0% adoption).

Every pipeline is independently authored. Without templates, changes to provisioning standards require updating each pipeline individually — leading to drift in pipeline design and inconsistent governance enforcement.

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Recommendation: Create 2–3 canonical IaCM templates:

- Terraform provisioning template (plan !' approval !' apply)
- Destroy template (approval gate !' destroy)
- Drift detection template (scheduled scan !' alert)

2.6 Module Registry

All 22 workspaces source Terraform/OpenTofu modules from the public HashiCorp registry only. No private module registry (Terraform Cloud Enterprise, JFrog Artifactory, or Harness-native) is configured.

Current state: Acceptable for simple use cases.

Risk: No version pinning enforcement, no internal module governance, no ability to restrict which modules teams can use.

Section III — Governance & Policy

3.1 OPA Policy Inventory

The account has a rich OPA policy library. 46 policies are authored and all are enabled. Critically, the IaCM-specific policies are well-developed:

Policy	Category
`open-tofu`	OpenTofu governance
`opentofu-s3-blockpublic`	S3 public access blocking
`opentofu-s3-bucket-logging`	S3 logging enforcement
`opentofu-s3-lifecycle-policy`	S3 lifecycle rules
`opentofu-s3-tags`	S3 tagging enforcement
`s3-deny-public-read-access`	S3 public read prevention
`s3-deny-without-lifecycle`	S3 lifecycle mandatory
`s3-enforce-versioning`	S3 versioning enforcement
`s3-opentofu-sse`	Server-side encryption
`s3-opentofu-versioning`	OpenTofu versioning
`s3-server-side-encryption`	SSE enforcement
`Terraform Plan Cost - Cost Estimate Increased`	Cost guardrail
`Terraform Plan Cost - Total Cost Estimate`	Cost budget enforcement

3.2 Policy Sets

Policy Set	Trigger	Enabled	Policies	Coverage
`Cost-Estimate-OPA`	afterTerraformPlan	Active	1	Account-wide
`OpenTofuS3`	afterTerraformPlan	Active	4	Account-wide
`New Policy Set - 10/13 - 20:11`	onRun	Disabled	1	Account-wide
`s3-deny-without-lifecycle`	afterTerraformPlan	Disabled	1	Account-wide
`s3-opa-policies`	afterTerraformPlan	Disabled	3	Account-wide
`s3-versioning-opentofu`	afterTerraformPlan	Disabled	0	Account-wide

3.3 Pipeline OPA Coverage

SUCCESS

100% of IaCM pipelines (15/15) are covered by OPA governance.

Both active policy sets are account-scoped, providing blanket enforcement across all projects and orgs.

WARNING

4 of 6 policy sets (67%) are disabled. These contain S3 lifecycle, versioning, and general OPA policies that are fully authored but not enforced:

- s3-deny-without-lifecycle — would enforce S3 lifecycle policies on all plans
- s3-opa-policies — contains 3 S3 compliance rules
- s3-versioning-opentofu — versioning enforcement for OpenTofu

Enabling these would immediately close 3 additional S3 compliance gaps without any authoring effort.

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RISK

All active policy sets trigger after TerraformPlan only — no onRun (shift-left) enforcement.

The New Policy Set - 10/13 - 20:11 has an onRun trigger but is disabled. Without pre-run checks, non-compliant runs are only blocked after a plan has already been generated — adding latency to the feedback loop.

Section IV — Areas of Improvement

Priority Matrix

Opportunity	Effort	Impact	Priority
Enable cost estimation on all workspaces	Low	High	**P1**
Enable Checkov on 13 pipelines	Medium	High	**P1**
Enable 4 disabled S3 OPA policy sets	Low	Medium-High	**P1**
Enable `onRun` OPA enforcement (shift-left)	Low	Medium	**P2**
Create IaCM pipeline templates	Medium	Medium	**P2**
Remediate 4 failed workspaces	Low	Medium	**P2**
Activate 7 unused IaCM-enabled projects	High	Medium	**P3**
Implement private module registry	High	Low-Medium	**P3**

Recommended Next Steps

ACTION

P1 — Enable Cost Estimation (this week)

Set cost_estimation_enabled: true on all 22 workspaces. Connect to Harness CCM for pre-apply cost diffs.

Activate the Terraform Plan Cost - Total Cost Estimate OPA policy set to enforce cost budgets at plan time.

Outcome: Full pre-apply cost visibility across all workspaces.

ACTION

P1 — Deploy Checkov to all provisioning pipelines (this sprint)

Add the IaCMCheckov step to the 13 pipelines currently missing it. Start with s3-prod-apply, tofus3-deploy, and s3-dev-destroy (production-impact pipelines first).

Outcome: Policy-as-code security scanning on 100% of provisioning runs.

ACTION

P1 — Enable 4 disabled OPA policy sets (1 hour effort)

The S3 compliance policies (s3-deny-without-lifecycle, s3-opa-policies, s3-versioning-opentofu) are fully authored. Simply toggle them enabled in the Harness Policy Management console.

Outcome: 7 additional S3 compliance rules enforced with zero authoring effort.

ACTION

P2 — Add onRun OPA enforcement (shift-left governance)

Enable New Policy Set - 10/13 - 20:11 (already has onRun trigger). Create additional pre-run checks for workspace configuration validation.

Outcome: Policy failures surface before plan generation — faster feedback, lower CI cost.

ACTION

P2 — Create 3 canonical IaCM pipeline templates

Author a Provision template (plan !' approval !' apply), a Destroy template (approval !' destroy), and a Drift Detection template. Migrate existing pipelines to use them.

Outcome: Consistent governance across all pipelines. Single-point updates to provisioning standards.

Appendix — Full Pipeline List

Pipeline	Project	Org	Checkov	OPA Coverage
Jira EC2 Provisioning	gketest	default		Cost-Estimate-OPA, OpenTofuS3
s3-dev-apply	gketest	default		Cost-Estimate-OPA, OpenTofuS3
Jira S3 Provisioning	gketest	default		Cost-Estimate-OPA, OpenTofuS3
IaCM Jira Automation	gketest	default		Cost-Estimate-OPA, OpenTofuS3
create-s3-site	gketest	default		Cost-Estimate-OPA, OpenTofuS3
s3-dev-destroy	gketest	default		Cost-Estimate-OPA, OpenTofuS3
s3-prod-apply	gketest	default		Cost-Estimate-OPA, OpenTofuS3
s3-dev-drift	gketest	default		Cost-Estimate-OPA, OpenTofuS3
tofus3-deploy	gketest	default		Cost-Estimate-OPA, OpenTofuS3
tofus3-destroy	gketest	default		Cost-Estimate-OPA, OpenTofuS3
iacm-provision-pipeline	chaos_project	default		Cost-Estimate-OPA, OpenTofuS3
iamcm-destroy-pipeline	chaos_project	default		Cost-Estimate-OPA, OpenTofuS3
ccm_connector_creation	chaos_project	default		Cost-Estimate-OPA, OpenTofuS3
Demo_Pipeline	chaos_project	default		Cost-Estimate-OPA, OpenTofuS3
Selective Approval Notification Pipeline	CCM-Demo	sam		Cost-Estimate-OPA, OpenTofuS3

Appendix — Full Workspace

Workspace	Project	Provisioner	Status	Cost Est.	Private Registry
DEV-ec2-SUP-13	gketest	terraform	active		
DEV-ec2-SUP-18	gketest	terraform	active		

DEV-Samriddha-Test-202025-ICM-10	gketest	terraform	failed
DEV-samtestfeb26-SUP-6	gketest	terraform	active
DEV-temsams3buck-ICM-11	gketest	terraform	active
null-null-null	gketest	terraform	failed
open tofu-s3	gketest	opentofu	active
open tofu-s3 destroy	gketest	opentofu	apply_needed
PROD-pleasetests3-ICM-6	gketest	terraform	active
PROD-temporarvsams3test-	gketest	terraform	active
QA-samriddhaes3-SUP-14	gketest	terraform	active
QA-samriddhatestccm-	gketest	terraform	active
QA-samriddtempors3-	gketest	terraform	active
QA-temp-s3-sam-test726-ICM-8	gketest	terraform	active
QA-test-bucket-SUP-17	gketest	terraform	failed
QA-test-s3-bukcet-ICM-7	gketest	terraform	failed
s3-dev	gketest	terraform	active
s3-new	gketest	opentofu	apply_needed
s3-prod	gketest	terraform	apply_needed
pga-aws	chaos_project	opentofu	—
test-idp	chaos_project	terraform	—
test-namespace-creation	chaos_project	terraform	—

Report generated by Harness IaC
Data source: Harness Platform A

